

AUSTIN-BERGSTROM, TX, USA

WMO: 722540

Lat: 30.183N

Lon: 97.680W

Elev: 146

StdP: 99.58

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13904

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.0	-1.2	-11.1	1.5	3.9	-8.6	1.8	5.8	11.2	8.6	10.4	9.9	2.7	0	0.442

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.5	37.9	23.5	36.9	23.6	35.8	23.7	25.9	31.8	25.6	31.4	25.3	30.9	4.1	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.7	20.1	27.2	24.3	19.6	27.0	24.0	19.2	26.9	81.0	31.9	79.3	31.5	77.8	30.9	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.8	8.7	7.9	-6.4	39.8	2.3	1.9	-8.1	41.1	-9.4	42.2	-10.7	43.3	-12.3	44.6		
(5)			-7.4	27.2	2.0	1.2	-8.8	28.1	-10.0	28.7	-11.1	29.4	-12.6	30.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.4	10.5	12.7	16.2	20.3	24.5	27.9	29.1	29.4	26.4	21.1	15.5	11.1
	DBStd	7.94	5.38	5.73	4.99	4.02	3.23	2.07	1.74	1.82	3.02	4.30	5.15	5.24
	HDD10.0	165	60	34	9	1	0	0	0	0	0	1	12	48
	HDD18.3	916	248	171	101	26	2	0	0	0	1	23	113	231
	CDD10.0	3974	75	109	202	309	451	537	592	601	491	346	177	84
	CDD18.3	1683	5	13	35	84	194	287	334	343	242	110	28	8
	CDH23.3	18617	33	96	239	709	1843	3318	4152	4337	2594	1048	205	42
CDH26.7	9111	3	19	43	206	750	1673	2282	2449	1275	372	35	3	
Wind	WSAvg	3.5	3.6	3.9	4.0	4.0	3.9	3.5	3.2	3.0	2.7	3.0	3.3	3.4
Precipitation	PrecAvg	898	71	51	67	70	107	75	66	51	73	116	76	68
	PrecMax	1574	255	195	224	170	358	288	247	197	190	542	259	330
	PrecMin	439	0	2	7	7	7	7	7	0	0	2	0	7
	PrecStd	298	61	46	45	42	74	61	52	50	47	120	67	69
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.4	29.2	29.9	33.3	35.5	38.5	38.9	39.4	38.4	33.8	29.8	26.7
		MCWB	16.6	17.7	19.3	21.5	23.4	23.4	23.3	23.7	22.6	22.5	20.7	18.5
	2%	DB	23.9	26.1	27.8	30.8	33.7	36.4	37.5	38.0	36.0	32.1	27.6	24.4
		MCWB	16.4	17.2	19.0	20.9	23.4	23.8	23.5	23.6	23.3	22.3	20.2	18.2
	5%	DB	21.9	23.9	26.1	28.9	32.3	35.1	36.5	37.0	34.4	30.5	25.8	22.6
		MCWB	16.1	17.0	18.6	20.6	23.2	24.0	23.7	23.7	23.3	21.6	19.2	17.4
	10%	DB	20.1	21.8	24.2	27.4	30.8	33.8	35.4	35.9	32.9	28.8	23.9	20.7
		MCWB	15.6	16.7	17.9	19.9	22.8	24.0	23.8	23.9	23.0	20.9	18.8	16.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.3	21.1	21.9	24.1	26.1	26.6	26.3	26.5	26.0	25.4	23.2	21.3
		MCDB	22.4	24.3	26.1	29.0	31.1	33.0	32.9	32.6	31.1	29.1	26.4	23.6
	2%	WB	19.3	20.1	21.1	23.1	25.2	25.8	25.7	25.8	25.5	24.6	22.2	20.3
		MCDB	21.7	23.0	24.4	27.7	30.2	31.8	31.9	31.8	30.5	28.3	25.0	22.8
	5%	WB	18.2	19.2	20.4	22.4	24.5	25.4	25.3	25.4	25.0	23.9	21.2	19.1
		MCDB	20.8	21.9	23.6	26.6	29.5	31.2	31.2	31.5	30.0	27.5	24.0	21.7
	10%	WB	16.5	18.0	19.6	21.7	23.9	25.0	25.0	25.1	24.5	23.1	20.1	17.5
		MCDB	19.4	20.9	22.9	25.5	28.7	30.5	30.8	31.1	29.4	26.6	23.2	20.0
Mean Daily Temperature Range	5% DB	MDBR	12.8	12.3	12.3	12.2	11.2	11.3	12.0	12.5	12.2	12.8	12.6	12.5
		MCDBR	15.3	14.7	13.4	13.3	12.5	13.0	13.7	14.2	13.9	13.8	13.8	14.6
		MCWBR	9.2	8.5	7.0	6.0	4.4	3.2	2.9	3.1	4.0	5.7	7.6	9.4
	5% WB	MCDBR	11.5	10.9	10.8	10.2	10.2	10.5	11.4	11.8	10.7	10.0	10.4	11.6
		MCWBR	8.4	7.5	6.8	5.4	4.3	3.4	3.1	3.1	3.7	4.9	7.1	8.9
Clear-Sky Solar Irradiance	taub		0.334	0.341	0.363	0.405	0.450	0.432	0.445	0.433	0.414	0.378	0.365	0.322
	taud		2.477	2.466	2.414	2.317	2.236	2.336	2.333	2.351	2.359	2.443	2.413	2.525
	Ebn at Noon		898	924	922	890	847	857	844	853	860	872	853	895
	Edh at Noon		92	101	114	130	142	128	127	124	119	102	97	84
All-Sky Solar Radiation	RadAvg		3.00	3.47	4.49	5.39	5.77	6.58	6.64	6.28	5.18	4.28	3.21	2.65
	RadStd		0.32	0.62	0.41	0.48	0.50	0.44	0.54	0.33	0.48	0.59	0.37	0.34

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+80

Nomenclature: See separate page